Lesson 25

American Inventions: The Television

Television has made it possible for people around the world to share in many different experiences – sports, music shows, films, and more.

Yet many people would find it difficult to name the inventor of the television. They might also be surprised to learn that the inventor came up with the idea when he was just fourteen years old.

Today, we explore the life of one of the great inventors of 20th century America: Philo Taylor Farnsworth.

A young boy

Philo Farnsworth was born in August 1906, near Indian Creek in the western state of Utah. The house he lived in for the first few years of his life had no electricity. But Philo read about electricity and quickly began to experiment with it.

One night, Philo read a magazine story about the idea of sending pictures and sound through the air. The story said some of the world's best scientists were working on the idea. It said these scientists were using special machines to try to make a kind of device to send pictures.

Fourteen-year-old Philo decided these famous scientists were wrong. He decided that mechanical devices would never work.

Philo believed such a device would have to be electronic. Philo knew electrons could be made to move very fast. All he would have to do was find a way to make electrons do the work.

Soon, Philo had an idea for such a receiver. It would trap light in a container and send the light on a line of electrons. Philo called it "light in a bottle."

Several days later, Philo told his teacher about a device that could capture pictures. He drew a plan for it and gave it to his teacher.

Philo's drawing seemed very simple. But it still clearly showed the information needed to build a television.

Philo's teacher was Justin Tolman. Many years later, Philo would credit Mr. Tolman with guiding his imagination and helping him open the doors of science.

Inventor

In September 1927, Philo turned on a device that was the first working television receiver. In another room was the first television camera. Philo had invented the special camera tube earlier that year.

Philo Farnsworth. (AP)

While the image produced on the receiver was not very clear, the device worked. Within a few months, Philo had found several people who wanted to invest money in his invention.

In August 1930, the United States government gave Philo patent documents. These documents aimed to protect his invention from being copied by others.

Still, he became involved in legal disputes with a powerful company at the time known as RCA. Philo won the disputes but faced many business and financial difficulties in his life.

He developed more than 100 devices that helped make modern television possible. He also developed early radar, invented the first electronic microscope, and worked on developing peaceful uses of atomic energy.

Philo Farnsworth died in March of 1971. He is considered one of the most important inventors of the 20th century.

I'm John Russell.

Paul Thompson wrote this story for VOA Learning English. John Russell adapted it.

Vocabulary

experiences - n. ประสบการณ์

inventor - n. นักประดิษฐ์

came up with - phrasal verb. คิดค้น, เสนอแนะ

experiment - v. ทดลอง

device - n. อุปกรณ์, เครื่องมือ

electronic - adj. อิเล็กทรอนิกส์

receiver - n. เครื่องรับสัญญาณ

capture - v. จับภาพ, บันทึกภาพ

credit - v. ยกความดีความชอบให้

patent - n. สิทธิบัตร

disputes - n. ข้อพิพาท, การโต้แย้ง

developed - v. พัฒนา, ประดิษฐ์

radar - n. เรดาร์

microscope - n. กล้องจุลทรรศน์

atomic energy - n. พลังงานปรมาณู

Reading Comprehension Quiz

1. What is mentioned as a benefit of television in the beginning of the article?

A) It connects people through live conversations.

B) It helps people learn new languages.

C) It allows sharing of various experiences like sports and music.

D) It provides instant news updates from around the world.

2. What might surprise many people about the inventor of television, according to the article?

A) That the inventor was from a different country.

B) That the inventor's name is not widely known.

C) That the inventor was very wealthy.

D) That the inventor was only fourteen when the idea came up.

3. Based on Paragraph 3, what is a common difficulty people face regarding the inventor of television?

A) Finding information about his life.

B) Naming the specific inventor.

C) Understanding the technology he used.

D) Recognizing his contributions to modern science.

4. What is the main purpose of Paragraph 4?

A) To introduce the main subject of the article.

B) To discuss the challenges faced by inventors.

C) To highlight the importance of 20th-century inventions.

D) To compare Philo Farnsworth with other inventors.

5. Where did Philo Farnsworth spend the first few years of his life?

A) In a large city with advanced technology.

B) In a house that had no electricity.

C) In a foreign country studying science.

D) In a school specializing in electrical engineering.

6. What was the subject of the magazine story that Philo read?

A) The history of electronic devices.

B) The challenges of modern communication.

C) The idea of transmitting pictures and sound wirelessly.

D) The lives of famous scientists.

7. What was Philo Farnsworth's initial reaction to the methods scientists were using to send pictures?

A) He agreed with their mechanical approach.

B) He believed their methods were incorrect.

C) He sought to collaborate with them.

D) He found their work inspiring but impractical.

8. What was Philo's core belief about how a device to send pictures should operate?

A) It should be powered by chemical reactions.

B) It must rely on mechanical components.

C) It should use electronic principles.

D) It needs to be a combination of mechanical and electronic parts.

9. What did Philo call his early idea for a receiver that would trap light and send it on a line of electrons?

A) Electron Beam Collector.

B) Light in a Bottle.

C) Image Transmission Device.

D) Visual Data Processor.

10. What did Philo do after developing his idea for a picture-capturing device?

A) He immediately sought investors for his invention.

B) He published his findings in a scientific journal.

C) He shared his plan and a drawing with his teacher.

D) He tried to build a prototype on his own.

11. How did Philo's drawing, despite its simplicity, prove significant?

A) It demonstrated his artistic talent.

B) It was a clear guide for building a television.

C) It showed his understanding of advanced physics.

D) It inspired other students to pursue science.

12. Who did Philo Farnsworth credit with helping him develop his scientific imagination?

A) His parents.

B) Other famous scientists.

C) Justin Tolman, his teacher.

D) Magazine articles he read.

13. When did Philo Farnsworth successfully operate his first working television receiver?

A) August 1930.

B) Early 1927.

C) September 1927.

D) March 1971.

14. What was the purpose of the patent documents granted to Philo in August 1930?

A) To recognize his scientific achievements globally.

B) To provide him with financial support.

C) To prevent others from copying his invention.

D) To allow him to sell his invention to the government.

15. Besides television-related devices, what other significant invention or area of work is attributed to Philo Farnsworth?

A) The internet.

B) Early radar technology.

C) The modern computer.

D) Space travel rockets.